

# Bachelor Thesis EDAR

*EDAR V370A: An Integrative Analysis of Evolutionary Conservation, Positive Selection, and Structural Tolerance*

EDAR VARIANT SET:

Group A: Clinically Described Variants (~27)

*From ClinVar (Pathogenic / Likely pathogenic). All variants are absent from 1000 Genomes Phase 3 (AF = 0), which is consistent with the rare pathogenic classification.*

| rsID         | Protein | Consequence             | ClinVar           | POS GRCh38  | AF EAS | AF AFR | AF EUR | AF SAS | AF AMR |
|--------------|---------|-------------------------|-------------------|-------------|--------|--------|--------|--------|--------|
| rs886039348  | W374*   | stop_gained             | Pathogenic/LP     | 108,897,133 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs557166582  | R98W    | missense_variant        | Pathogenic/LP     | 108,929,262 | 0.000  | 0.000  | 0.000  | 0.001  | 0.000  |
| rs121908450  | R89H    | missense_variant        | Pathogenic/LP     | 108,929,288 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs550414401  | A26V    | missense_variant        | Pathogenic/LP     | 108,930,217 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs886039564  | R358Q   | missense_variant        | Pathogenic/LP     | 108,897,181 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs2470612697 | C428R   | missense_variant        | Pathogenic/LP     | 108,896,972 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs121908453  | R420Q   | missense_variant        | Pathogenic/LP     | 108,896,995 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs2470612633 | E433*   | stop_gained             | Pathogenic        | 108,896,957 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs773132518  | R25*    | stop_gained             | Pathogenic        | 108,930,221 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs2105390253 | —       | splice_acceptor_variant | Pathogenic        | 108,906,369 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs2470612948 | Y406*   | stop_gained             | Pathogenic        | 108,897,036 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs121908452  | R358*   | stop_gained             | Pathogenic        | 108,897,182 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs1696805876 | K327*   | stop_gained             | Pathogenic        | 108,906,353 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs121908456  | E354*   | stop_gained             | Pathogenic        | 108,897,194 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs1432041144 | E311*   | stop_gained             | Pathogenic        | 108,907,892 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs199544410  | C301*   | stop_gained             | Pathogenic        | 108,907,920 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs773885029  | W434G   | missense_variant        | Pathogenic        | 108,896,954 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs1696608702 | I431T   | missense_variant        | Pathogenic        | 108,896,962 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs368841777  | —       | splice_donor_variant    | Pathogenic        | 108,923,367 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs1574368295 | —       | splice_donor_variant    | Pathogenic        | 108,906,306 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs1553444895 | —       | splice_donor_variant    | Pathogenic        | 108,906,307 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs797044437  | —       | splice_donor_variant    | Pathogenic        | 108,910,459 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs1553445945 | —       | splice_donor_variant    | Pathogenic        | 108,912,677 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs797044435  | —       | splice_donor_variant    | Pathogenic        | 108,930,963 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs2470708774 | Y136*   | stop_gained             | Likely pathogenic | 108,923,402 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs749688157  | C31*    | stop_gained             | Likely pathogenic | 108,930,201 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| rs886041005  | C428*   | stop_gained             | Likely pathogenic | 108,896,970 | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |

Group B: Neutral Common Variants (~6)  
From gnomAD v4.1.1. with AF > 1% in at least one superpopulation, AF < 85% globally.

| rsID        | Protein | Consequence        | POS GRCh38  | AF EAS | AF AFR | AF EUR | AF SAS | AF AMR |
|-------------|---------|--------------------|-------------|--------|--------|--------|--------|--------|
| rs61761321  | M107V   | missense_variant   | 108,929,235 | 0.090  | 0.001  | 0.000  | 0.002  | 0.000  |
| rs114808659 | G239R   | missense_variant   | 108,910,887 | 0.000  | 0.083  | 0.000  | 0.000  | 0.004  |
| rs79648056  | S274S   | synonymous_variant | 108,908,001 | 0.000  | 0.082  | 0.000  | 0.000  | 0.004  |
| rs79798733  | V246M   | missense_variant   | 108,910,527 | 0.000  | 0.029  | 0.000  | 0.000  | 0.000  |
| rs3749099   | P290P   | synonymous_variant | 108,907,953 | 0.014  | 0.005  | 0.009  | 0.029  | 0.004  |
| rs146567337 | S380R   | missense_variant   | 108,897,116 | 0.031  | 0.000  | 0.000  | 0.001  | 0.000  |

Group C: Positive Control  
V370A (rs3827760) selected as primary positive control  
And two Linkage Disequilibrium partners identified by plink2 with r² > 0.88

| rsID       | Protein | Consequence      | ClinVar | POS GRCh38  | AF EAS | AF AFR | AF EUR | AF SAS | AF AMR | r² V370A |
|------------|---------|------------------|---------|-------------|--------|--------|--------|--------|--------|----------|
| rs3827760  | V370A   | missense_variant | —       | 108,896,851 | 0.873  | 0.003  | 0.011  | 0.013  | 0.391  | 1.000    |
| rs4676213  | —       | intron_variant   | —       | 108,866,545 | 0.859  | 0.003  | 0.011  | 0.014  | —      | 0.897    |
| rs72627476 | —       | intron_variant   | —       | 108,834,368 | 0.862  | 0.003  | 0.011  | 0.012  | —      | 0.886    |

